



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 18: General Equilibrium II

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ICS2523 Microeconomics

Lecture 18

Last lecture, we finished competitive markets

- Finished welfare analysis
- We started with **General Equilibrium**:
- General Equilibrium with Two Goods $\rightarrow$ Edgeworth Box

In today's lecture, we will finish **General Equilibrium**.

- See another example of Edgeworth Box
- Study the Production Possibility Frontier (PPF)

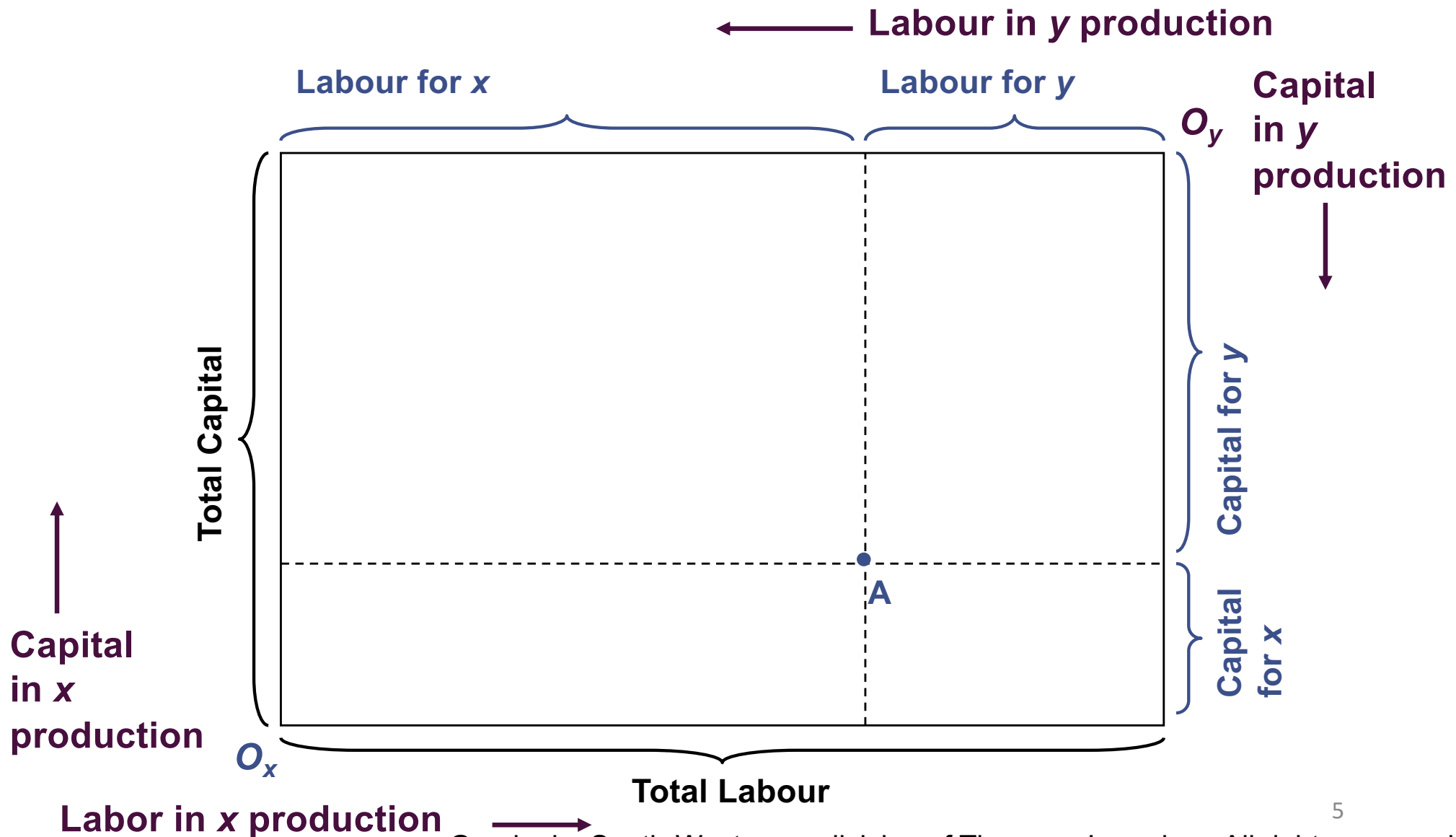
Model of General Equilibrium with Two Goods

Competitive Markets

9. General Equilibrium with Two Goods: Edgeworth Box (review)

- Two goods $\rightarrow x$ and y
- General equilibrium demand:
 - Assume that all individuals have identical preferences, represented by an indifference curve map defined over quantities of x and y $\rightarrow$ show how individuals rank consumption bundles containing both goods
- General equilibrium supply
 - Use the production possibility curve $\rightarrow$ Edgeworth box diagram
 - We must assume that there are fixed amounts of capital and labour inputs that must be allocated to the production of these two goods
 - $\rightarrow$ The possible allocation of these inputs is shown in an Edgeworth box diagram, with dimensions given by the total amounts of capital and labour available

Edgeworth Box Diagram (review)



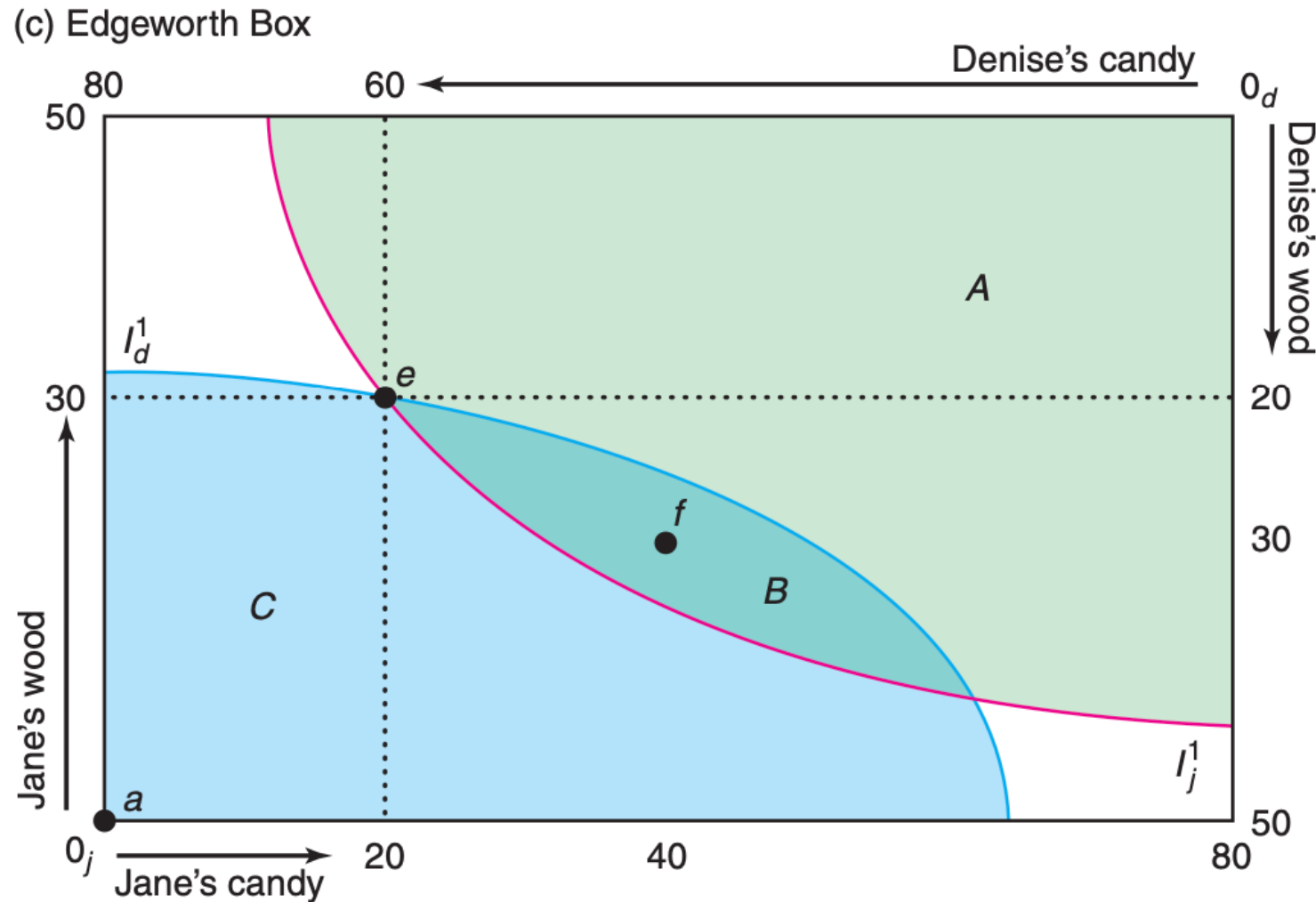
Example: Edgeworth box in Trading with two people II

Jane and Denise are neighbours, and a storm isolates them. They must trade with each other or consume only what they have at hand.

In total, they have 50 cords of firewood and 80 candy bars (and no way of producing more of either good).

Jane's **endowment** (her initial allocation of goods) is 30 cords of firewood and 20 candy bars. Thus, Denise's endowment is 20 cords of firewood and 60 candy bars.

Example: Edgeworth box in Trading with two people II



Example: Edgeworth box in Trading with two people II

Should Jane and Denise trade?

It will depend on their tastes → indifference curves

We will make 4 **assumptions**:

1. Utility maximisation
2. Convex indifference curves
3. Nonsatiation
4. No interdependence (neither person's utility depends on the other's consumption, and neither person's consumption harms the other person).

Example: Edgeworth box in Trading with two people II

Should Jane and Denise trade?

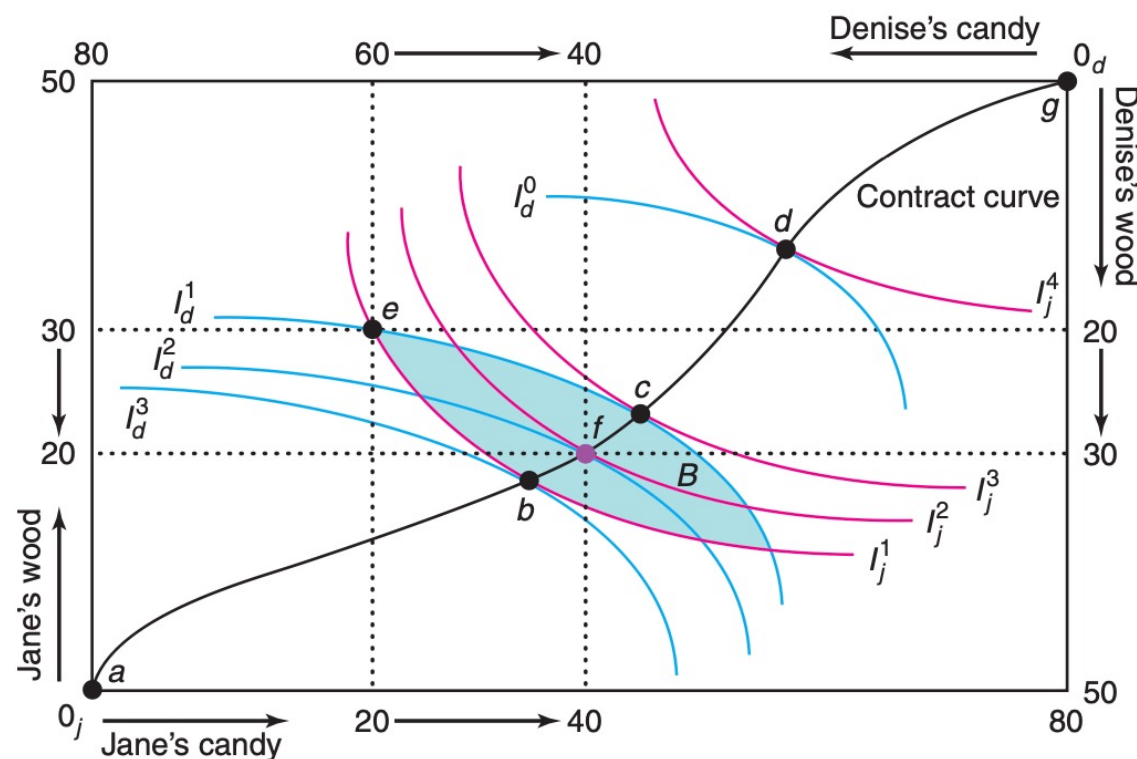
Suppose that they trade, reallocating goods from Bundle e to Bundle f

→ Jane gives up 10 cords of wood for 20 more candy bars, Denise gives up 20 candy bars for 10 cords of wood.

Both gain from such a trade (they reach higher indifference curves).

Example: Edgeworth box in Trading with two people II

Standing in bundle f , do Jane and Denise want to make further trades?



	Endowment, e		Trade		New Allocation, f	
	Wood	Candy	Wood	Candy	Wood	Candy
Jane	30	20	-10	+20	20	40
Denise	20	60	+10	-20	30	40

Example: Edgeworth box in Trading with two people II

No further trade is possible at a bundle like f because Jane's marginal rate of substitution (the slope of her indifference curve), MRS_j , between wood and candy equals Denise's marginal rate of substitution, MRS_d .

Jane's MRS_j is $-1/2$: She is willing to trade one cord of wood for two candy bars.

Because Denise's indifference curve is tangent to Jane's, Denise's MRS_d must also be $-1/2$. They can't agree on further trades when they both want to trade wood for candy at the same rate.

Example: Edgeworth box in Trading with two people II

In contrast, at a bundle such as e where their indifference curves are not tangent, MRS_j does not equal MRS_d .

Denise's MRS_d is $-1/3$, and Jane's MRS_j is -2 .

Denise is willing to give up one cord of wood for three more candy bars or to sacrifice three candy bars for one more cord of wood. If Denise offers Jane three candy bars for one cord of wood, Jane will accept because she is willing to give up two cords of wood for one candy bar. This example illustrates that trades are possible where indifference curves intersect because marginal rates of substitution are unequal.

Example: Edgeworth box in Trading with two people II

To summarize, we can make four equivalent statements about allocation f :

1. The indifference curves of the two parties are tangent at f .
2. The parties' marginal rates of substitution are equal at f .
3. No further mutually beneficial trades are possible at f .
4. The allocation at f is **Pareto efficient**: One party cannot be better off without harming the other.

* We will study Pareto efficiency next lecture

15. Economic efficiency

Definition: **Pareto efficient allocation**

An allocation of resources is Pareto efficient if it is not possible (through further reallocations) to make one person better-off without making someone else worse-off

Vilfredo Pareto
(1848-1923)



Example: Edgeworth box in Trading with two people II

Indifference curves are also tangent at bundles b , c , and d , so these allocations, like f , are Pareto efficient.

By connecting all such bundles, we draw the **contract curve**: the set of all Pareto-efficient bundles.

This name is given because only at these points are the parties unwilling to engage in further trades or contracts—these allocations are the final contracts. A move from any bundle on the contract curve would harm at least one person.

Example: Edgeworth box in Trading with two people II

We can use calculus to derive the contract curve:

We want to specify the conditions where we make one individual as well off as possible, without harming the other person.

Let Denise's utility function be $U_d(q_{d1}, q_{d2})$, where q_{d1} is the amount of candy and q_{d2} is the amount of Wood belonging to Denise.

Similarly, Jane's utility function is $U_j(q_{j1}, q_{j2})$,

Example: Edgeworth box in Trading with two people II

We want to find the bundle that maximises Jane's well being, $U_j(q_{j1}, q_{j2})$, given that we hold Denise's utility constant at $U_d(q_{d1}, q_{d2}) = \overline{U_d}$,

Thus, we want to:

$$\text{Max } U_j(q_{j1}, q_{j2})$$

$$\text{Subject to: } U_d(q_1 - q_{j1}, q_2 - q_{j2}) = \overline{U_d} \quad (\lambda)$$

$$q_1 = q_{j1} + q_{d1}$$

$$q_2 = q_{j2} + q_{d2}$$

Example: Edgeworth box in Trading with two people II

We can write the Lagrangian corresponding to the maximum problem as:

$$\mathcal{L} = U_j(q_{j1}, q_{j2}) + \lambda(U_d(q_1 - q_{j1}, q_{d2} - q_{j2}) - \overline{U_d})$$

The first order conditions are:

$$\frac{\partial \mathcal{L}}{\partial q_{j1}} = \frac{\partial U_j}{\partial q_{j1}} - \lambda \frac{\partial U_d}{\partial q_{j1}} = 0 \quad eq. 1$$

$$\frac{\partial \mathcal{L}}{\partial q_{j2}} = \frac{\partial U_j}{\partial q_{j2}} - \lambda \frac{\partial U_d}{\partial q_{j2}} = 0 \quad eq. 2$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = U_d(q_1 - q_{j1}, q_{d2} - q_{j2}) - \overline{U_d} = 0$$

Example: Edgeworth box in Trading with two people II

Equating the right-hand side of Eq.1 and Eq. 2, we find that

$$MRS_j = \frac{\frac{\partial U_j}{\partial q_{j1}}}{\frac{\partial U_j}{\partial q_{j2}}} = \frac{MU_{j1}}{MU_{j2}} = \frac{\frac{\partial U_d}{\partial q_{d1}}}{\frac{\partial U_d}{\partial q_{d2}}} = \frac{MU_{d1}}{MU_{d2}} = MRS_d$$

Example: Edgeworth box in Trading with two people II

Competitive Equilibrium

If the market has many people with tastes and endowments like Jane's and many people with tastes and endowments like Denise's, each person is a price taker in the two goods. We can use an Edgeworth box to examine how such price takers would trade.

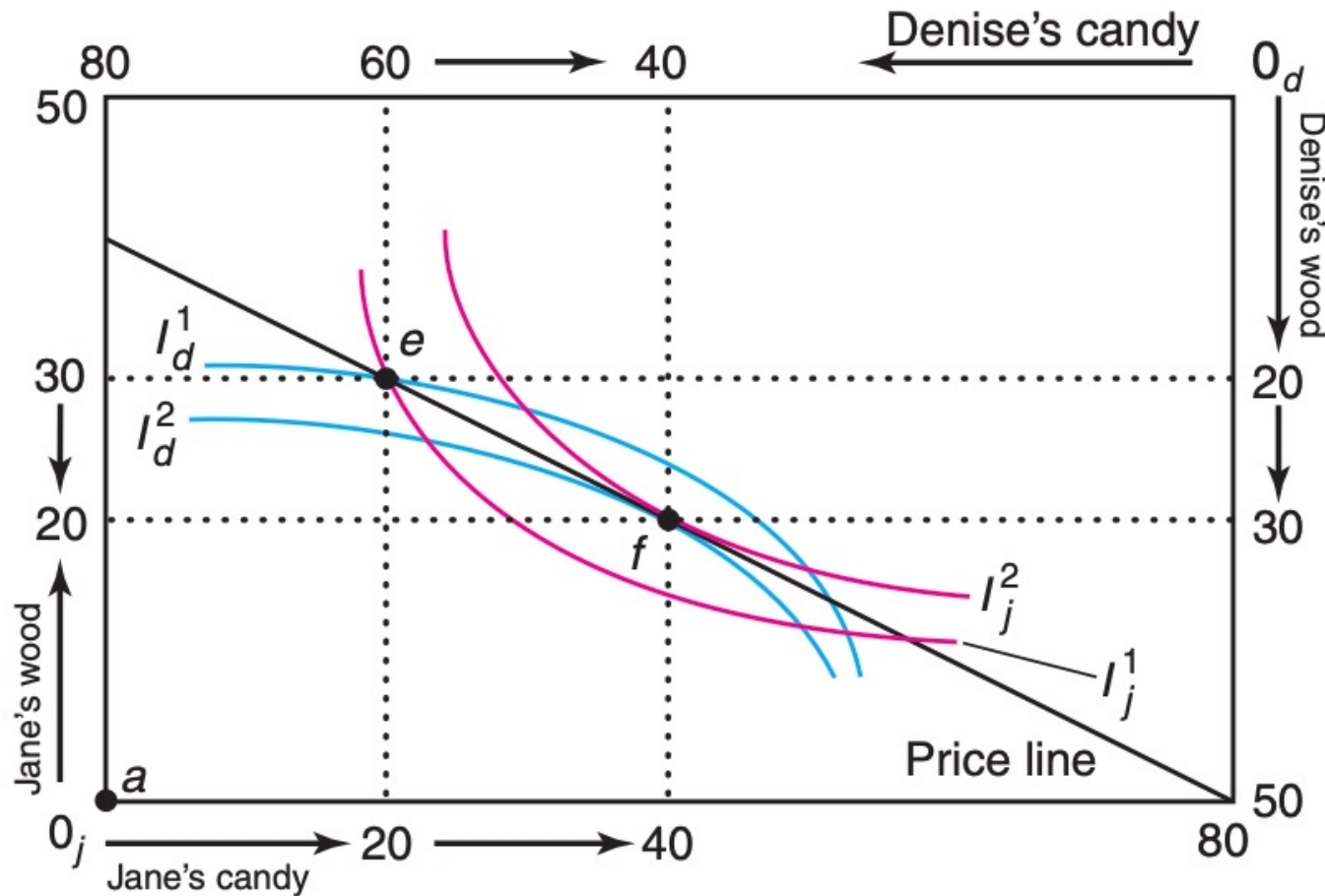
If the price of a cord of wood, P_w , is \$2, and the price of a candy bar, P_c , is \$1, then a candy bar costs half as much as a cord of wood:

$$\frac{P_c}{P_w} = \frac{1}{2}$$

An individual can sell one cord of wood and use that money to buy two candy bars.

Example: Edgeworth box in Trading with two people II

(a) Price Line That Leads to a Competitive Equilibrium

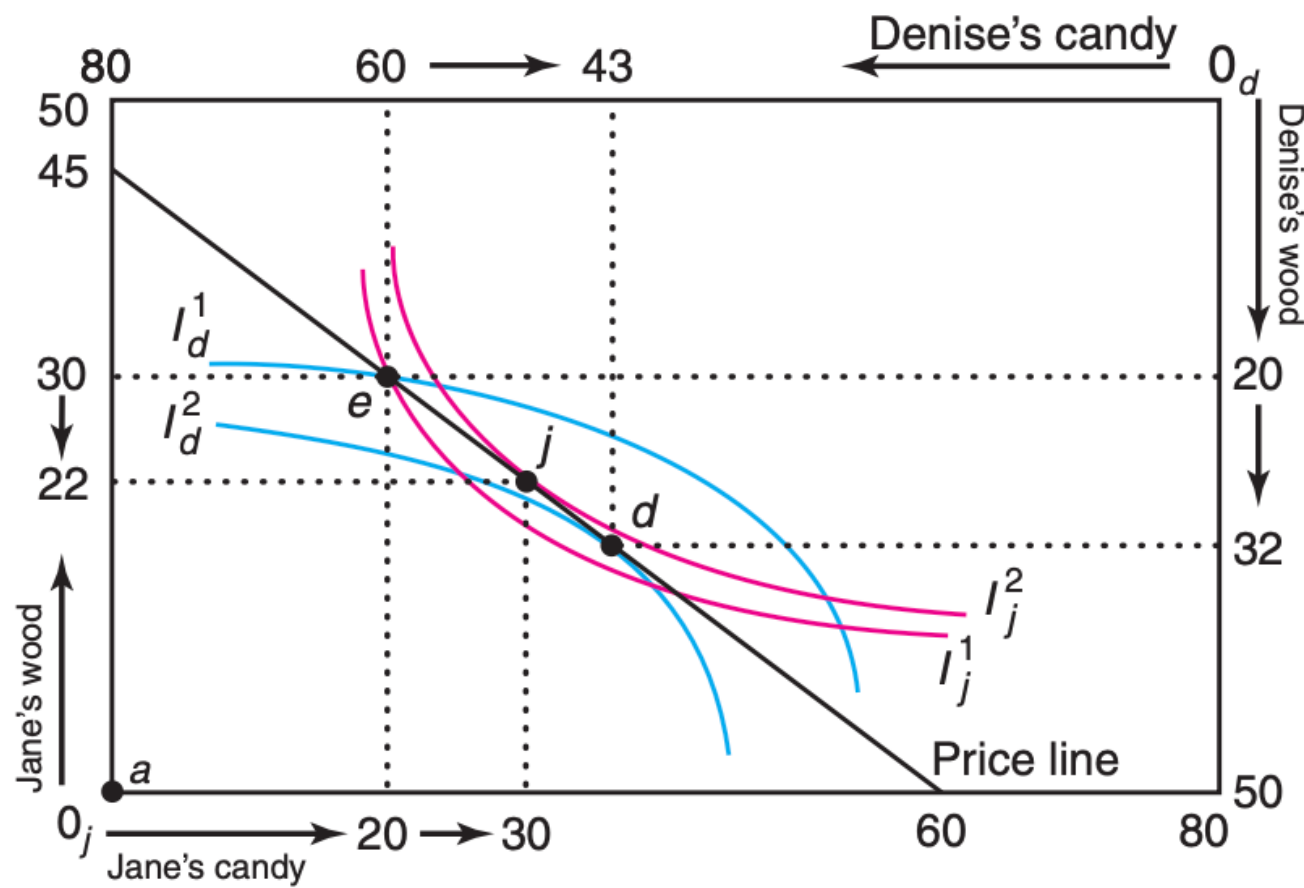


Perloff (2015)

Initial endowment is e and $\frac{P_c}{P_w} = \frac{1}{2}$

Example: Edgeworth box in Trading with two people II

(b) Prices That Do Not Lead to a Competitive Equilibrium



Example: Edgeworth box in Trading with two people II

The Efficiency of Competition

In a competitive equilibrium, the indifference curves of both types of consumers are tangent at the same bundle on the price line. As a result, the slope (MRS) of each person's indifference curve equals the slope of the price line, so the slopes of the indifference curves are equal:

$$MRS_j = \frac{MU_{j1}}{MU_{j2}} = \frac{P_1}{P_2} = \frac{MU_{d1}}{MU_{d2}} = MRS_d$$

UNTIL HERE FOR TEST 3

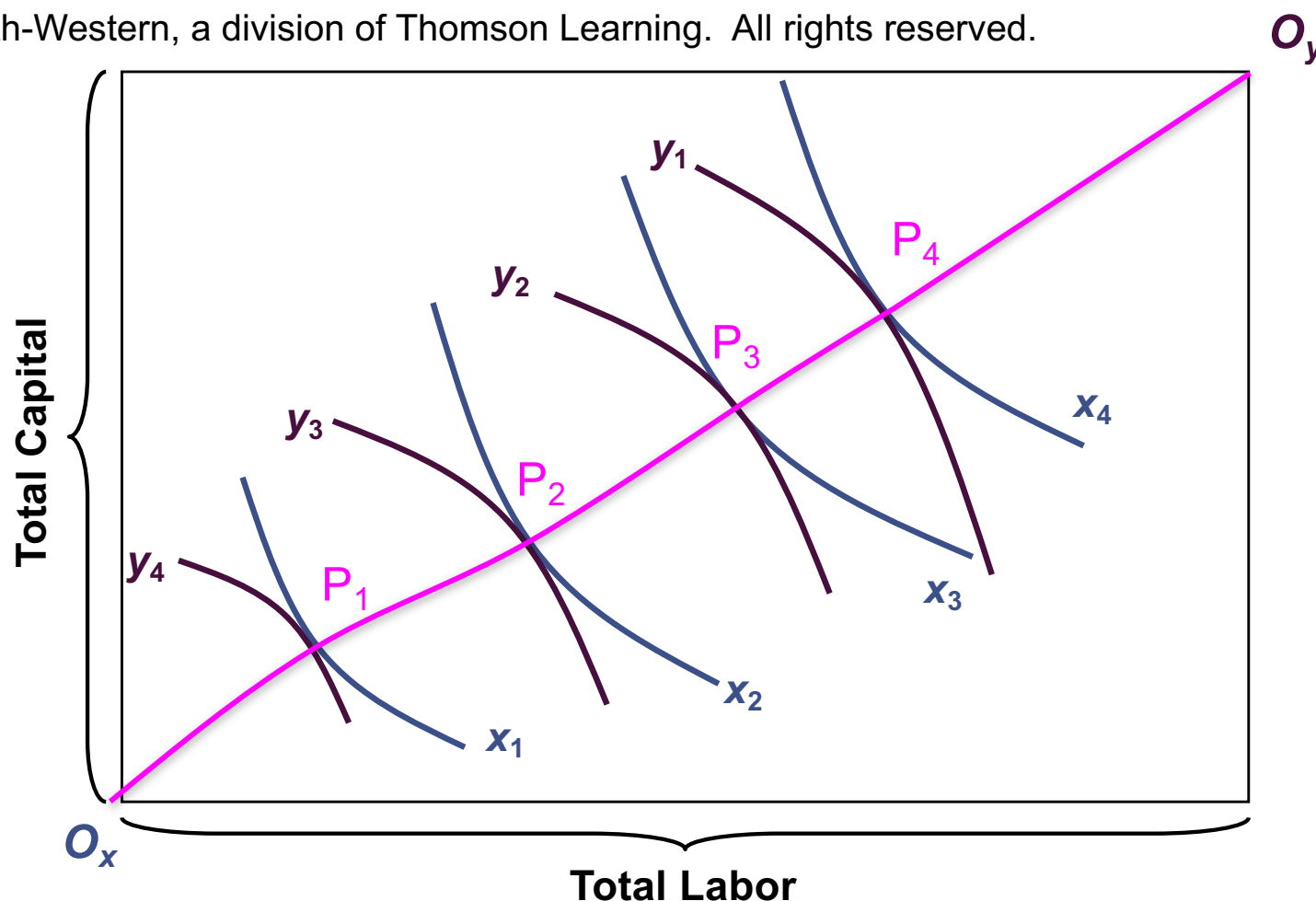
Production Possibility Frontier

Competitive Markets

9. Edgeworth Box Diagram (review)

Efficient locations are those such as P_1 , P_2 , P_3 and P_4
(isoquants are tangent to one another)

Graphs by South-Western, a division of Thomson Learning. All rights reserved.

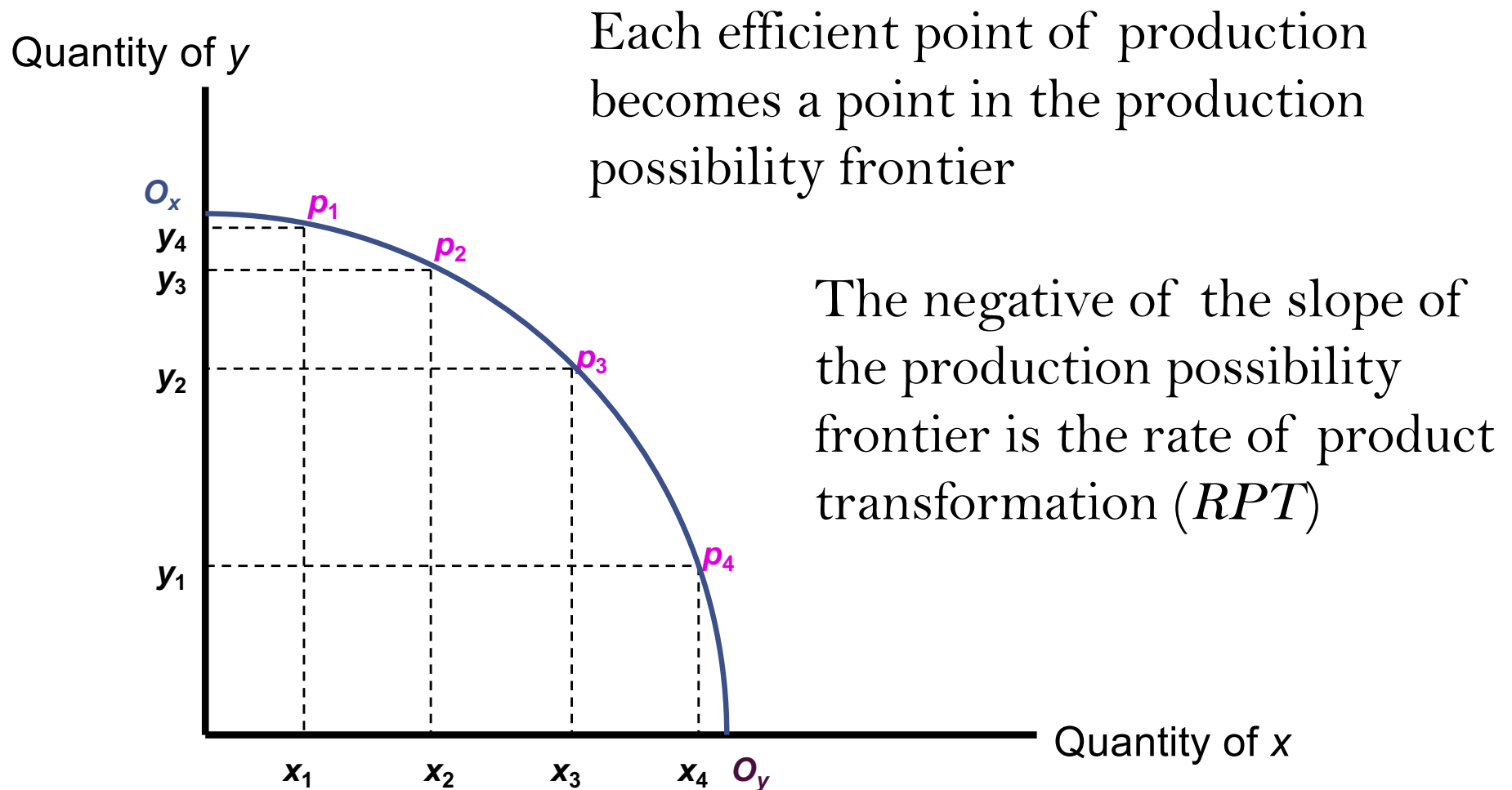


10. Production Possibility Frontier

Production possibility frontier (PPF)

- Efficient locations are those such as P_1 , P_2 , P_3 and P_4 (isoquants are tangent to one another)
- The locus of efficient points shows the maximum output of y that can be produced for any level of x
- We can use this information to construct a production possibility frontier (PPF)
- PPF $\rightarrow$ shows the alternative outputs of x and y that can be produced with the fixed capital and labour inputs that are employed efficiently

10. Production Possibility Frontier



10. Production Possibility Frontier

Rate of Product Transformation (RPT)

- **The slope of the PPF** $\rightarrow$ shows how output x can be substituted by y output when total resources are held constant.
- $RPT \text{ (of } x \text{ and } y) = - (\text{slope of PPF}) = -dy/dx$

$$RPT = \frac{MC_x}{MC_y}$$

10. Production Possibility Frontier

Why? → Intuition

Proof: the total differential of the cost function along the PPF is:

$$dC = \frac{\partial C}{\partial x} \cdot dx + \frac{\partial C}{\partial y} \cdot dy = 0$$

$$RPT = -\frac{dy}{dx} = \frac{\partial C / \partial x}{\partial C / \partial y} = \frac{MC_x}{MC_y}$$

Shape of the PPF

RPT increases as we move from O_x to O_y → **concave**

Activity 1

Juan can produce three chocolate bars or six paper notebooks per day. By dividing his time between these two activities, he can make various combinations of the two goods (any linear combination of notebooks and chocolates daily).

Denise can produce up to three paper notebooks or six chocolate bars each day (or any linear combination), and Hugo can produce five paper notebooks or five chocolate bars in a day (or any linear combination).

- a) Graph each person's individual production possibility frontier (PPF) using the horizontal axis for the production of chocolate bars and the vertical axis for the notebooks of paper.
- b) If they decide to collaborate, draw the joint PPF.

10. Production Possibility Frontier

- **Diminishing returns**

- Assumption: both goods are produced under diminishing returns → ↑ output x will raise its marginal cost, whereas decreasing output y will reduce its marginal cost.

- **Specialised inputs**

- If some inputs are more “suited” for x production than for y production → concave shape could also be explained.

- **Differing factor intensities**

- If inputs are homogeneous and the production function exhibits constant returns to scale, PPF will be concave if goods use inputs in different proportions.

- **Opportunity cost and supply**

- PPF curve demonstrates that there are many efficient combinations of the two goods and that producing more of one good implies cutting back on the production of the other good → **opportunity cost**

Activity 2

Suppose that the production of x and y depends only on labour input (total labour supply is 100) and that production functions (with decreasing returns) for these goods are:

$$\begin{aligned}x &= f(l_x) = l_x^{0.5} \\ y &= f(l_y) = l_y^{0.5}\end{aligned}$$

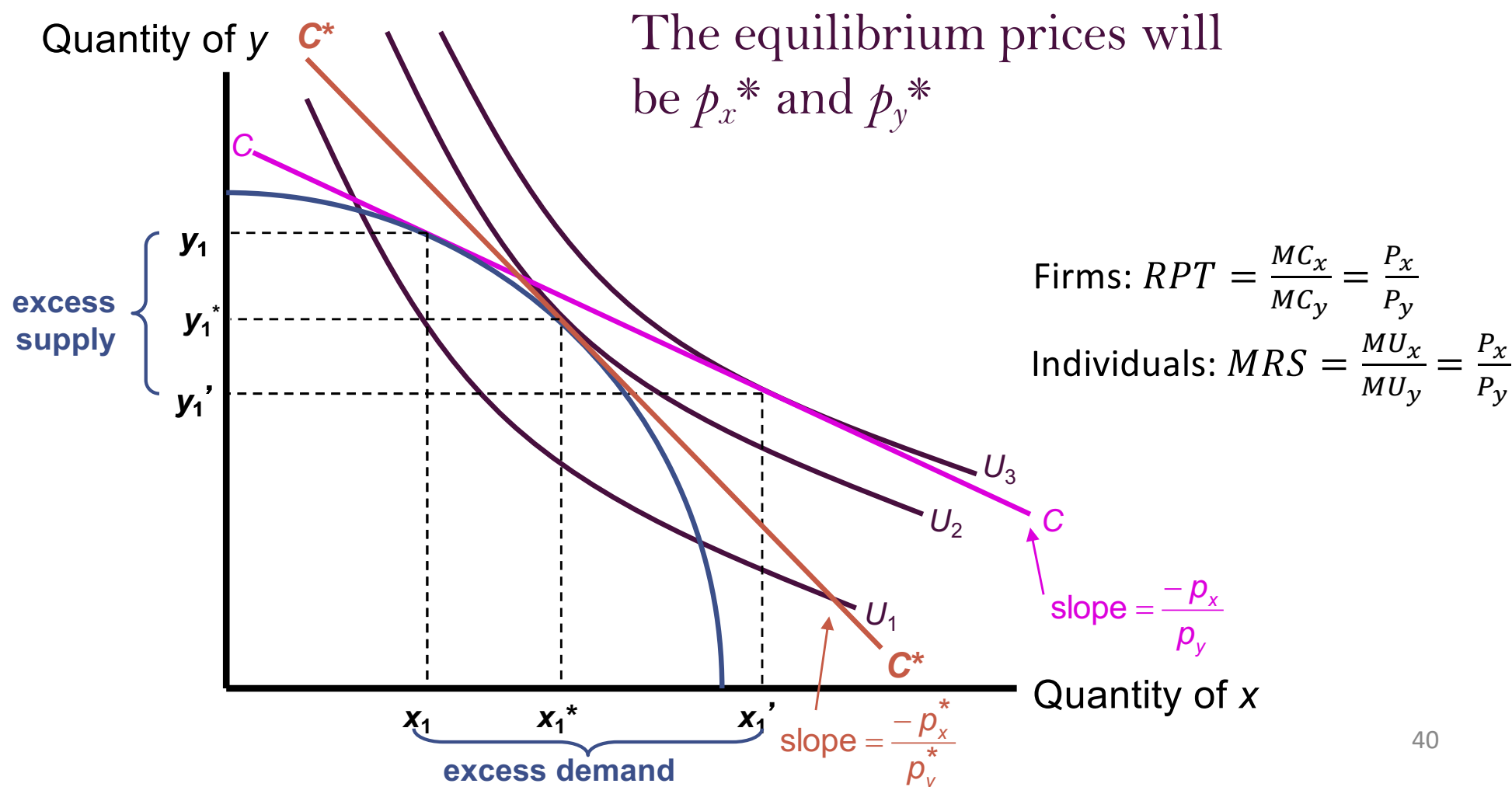
Find a mathematical expression for the PPF and for RPT

Determination of Equilibrium Prices

Competitive Markets

11. Determination of Equilibrium Prices

Individual's preferences are represented by the set of indifference curves.



11. Determination of Equilibrium Prices

Changes in demand

What would happen if preferences were to shift towards good x?

- Think about P_x/P_y
- Then, think about what might happen to x and y (consumption and production)

Changes in supply

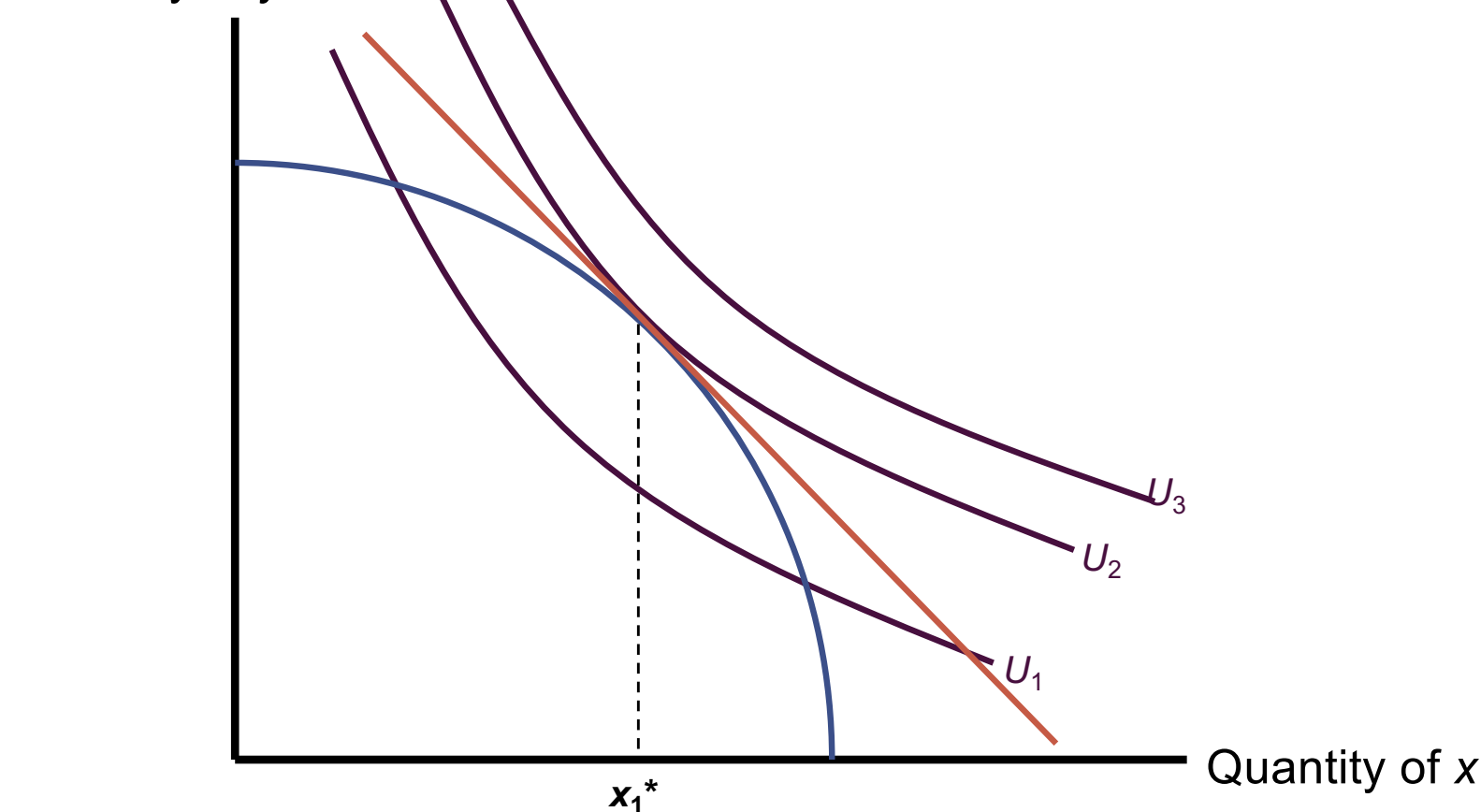
What will happen if there is technical progress in x production?

- Will shift PPF outward
- This will lower the relative price of x and increase the quantity of x consumed (if x is a normal good)
- What happens to y?

11. Determination of Equilibrium Prices

Comparative statics analysis

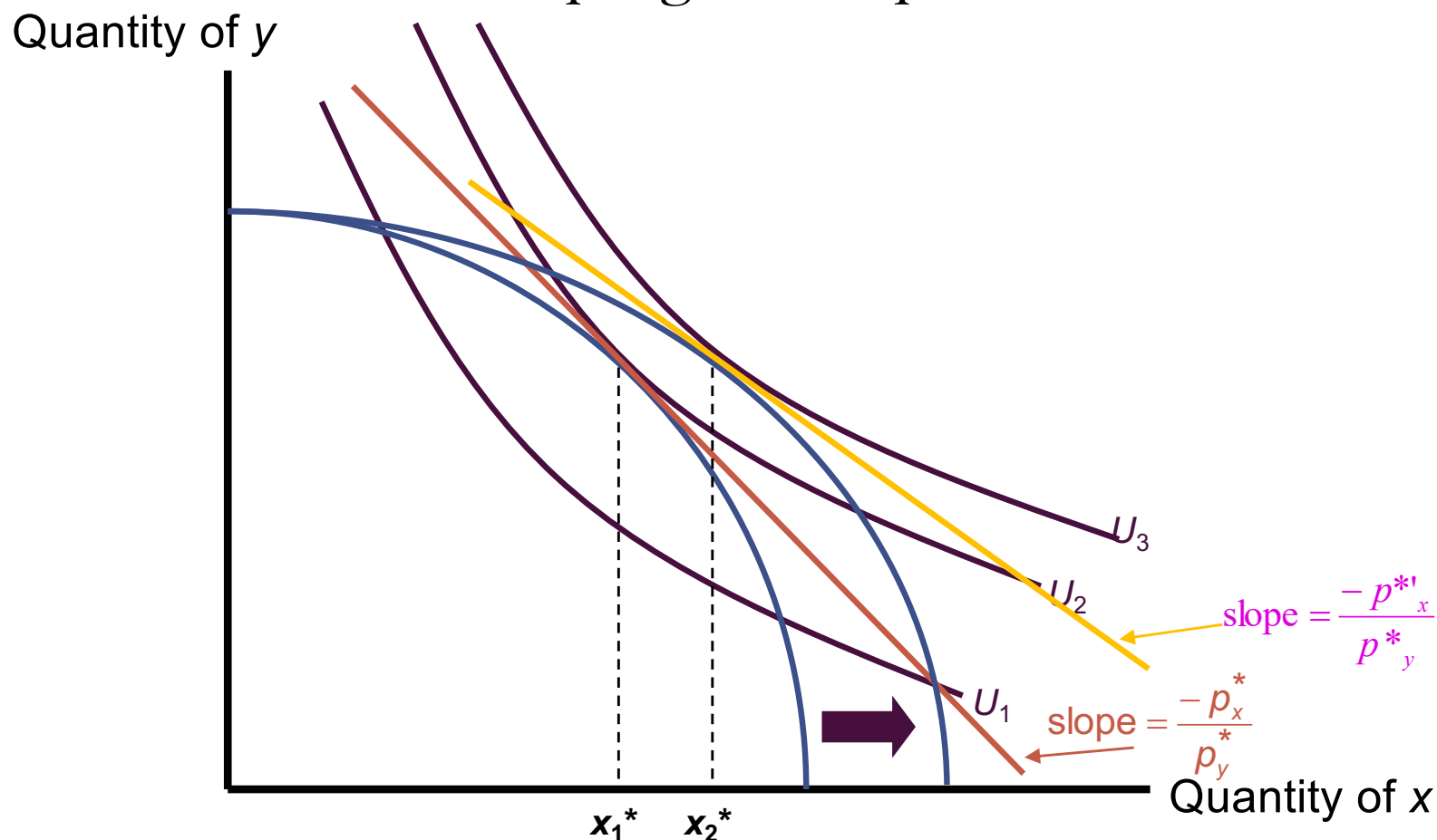
Effect of technical progress in production of x



11. Determination of Equilibrium Prices

Comparative statics analysis

Effect of technical progress in production of x



Existence of General Equilibrium Prices

Competitive Markets

12. Existence of General Equilibrium Prices

So far, we have assumed that competitive markets can reach an equilibrium. But, that equilibrium is not ensured (given the assumptions we have made)

A simple mathematical model: Suppose there are:

- n goods, distributed among the individuals in the society
- S_i for $(i=1, \dots, n) \rightarrow$ total supply available
- P_i for $(i=1, \dots, n) \rightarrow$ price of good i
- Total demand for good i depends on all the prices: $D_i(P_1, \dots, P_n)$ for $(i=1, \dots, n)$

Walras' problem can be stated as:

**Does there exist *an equilibrium set of prices* such that:
 $D_i(P_1^*, \dots, P_n^*) = S_i$, for all values of i ?**

12. Existence of General Equilibrium Prices

We can define *Excess Demand functions*:

$$ED_i(P_1, \dots, P_n) = D_i(P_1, \dots, P_n) - S_i, \text{ for } i=1, \dots, n$$

Equilibrium conditions can be rewritten as:

$$ED_i(P_1, \dots, P_n) = D_i(P_1, \dots, P_n) - S_i = 0, \text{ for } i=1, \dots, n$$

Remember:

Demand functions (and also the excess of demand functions) are homogeneous of grade 0 in prices (incomes are wages now, also a price)

Demand functions (and also the excess of demand functions) are continuous.

Walras' law: n excess demand functions are not independent of each other

$$\sum_{i=1}^n P_i \cdot ED_i(P_1, \dots, P_n) = 0$$

Attention: Walras law holds for any set of prices (not only equilibrium)

12. Existence of General Equilibrium Prices

Walras' law $\rightarrow$ total value of excess demand is zero at any set of prices

Two goods (A and B) and two individuals (Smith and Jones) in society

Smith's budget constraint:

$$P_a D_a^s + P_b D_b^s = P_a S_a^s + P_b S_b^s$$

$$\rightarrow P_a (D_a^s - S_a^s) + P_b (D_b^s - S_b^s) = 0 \rightarrow P_a E D_a^s + P_b E D_b^s = 0$$

Similarly, for Jones: $P_a E D_a^J + P_b E D_b^J = 0$

So, $P_a (E D_a^s + E D_a^J) + P_b (E D_b^s + E D_b^J) = 0$

$$\sum_{i=1}^n P_i \cdot E D_i(P_1, \dots, P_n) = 0$$

12. Existence of General Equilibrium Prices

Walras' proof of the existence of equilibrium prices

Walras tried to establish that the n equilibrium conditions
 $(ED_i(P_1, \dots, P_n) = D_i(P_1, \dots, P_n) - S_i = 0, \text{ for } i=1, \dots, n)$
where sufficient

It is $(n-1)$ independent equations for $(n-1)$ unknown relative prices

But, solving this equations system is not simple: equations are not always linear, and prices must be nonnegative

→ Walras developed a tedious proof that involved solving for equilibrium prices in a series of successive approximations

12. Existence of General Equilibrium Prices

Walras proof:

- Start with an initial set of prices (arbitrary)
- Holding the other $(n-1)$ prices constant, find the equilibrium price in the market for good 1
- Holding this price and the other $(n-2)$ prices constant, solve for the equilibrium price in the market for good 2
- Continue with proof until a complete set of provisional relative prices has been calculated, and continue to iterate until a reasonable approximation to a set of equilibrium is achieved.
- The system of equations is simultaneous. When one price changes, the price previously calculated need no longer be an equilibrium price

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.